THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 5: **Exploring Forces** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. When two forces F1 = 5 N (right) and F2 = 8 N (left) act on a body, a third force of what size and direction makes the net force zero? 1

(a) 13 N to the left **(b)** 13 N to the right
(c) 3 N to the left **(d)** 3 N to the right

2. A pair of identical rods, A and B, are charged by rubbing against different materials. They repel each other. This means: 1

(a) both have the same sign of charge **(b)** one is neutral
(c) both have opposite charges **(d)** both are magnets

3. Assertion (A): Lightning conductors have pointed ends and earth connections. Reason (R): They allow charge to flow safely to the ground. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. A charged glass rod attracts a suspended light ball. Which conclusion is certain? 1

(a) the ball is positively charged
(b) the ball is heavier
(c) the ball is either negatively charged or neutral
(d) the ball is a magnet

5. A boy of weight 400 N stands on a floor. The reaction from the floor is: 1

(a) 400 N downward **(b)** 0 N **(c)** 400 N upward **(d)** 200 N upward

6. Assertion (A): A charged rod attracts a neutral piece of paper. Reason (R): Neutral paper contains no charges at all. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. A charged rod repels a suspended ball. Which conclusion is certain? 1

(a) the ball has the same type of charge **(b)** the ball has opposite charge
(c) the ball is neutral **(d)** the ball is a magnet

8. Which of the following will increase the friction between a box and the floor? 1

(a) using a smoother box **(b)** putting a heavy load on the box
(c) oiling the floor **(d)** polishing the floor

9. **Assertion (A):** Two equal and opposite forces acting on a body always make it move. 1
Reason (R): Balanced forces have zero net force.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A rubbed plastic comb attracts small bits of paper. 1
Reason (R): Gravitational force acts between the comb and the paper.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does Earth's gravitational force not change the mass of a body? 2

12. A force of 10 N acts on a stone to the east and 10 N to the west. Predict the effect on the stone and the force on a hand that holds it. 2

13. An electroscope is charged by a rod. How does the divergence of leaves tell about the amount of charge? 2

14. Why do trucks carrying petrol have a metal chain touching the ground? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Two identical balloons are rubbed with wool and hung close together. What happens? What if one is rubbed with wool and the other with a glass rod? 3

16. A negatively charged rod touches an electroscope. Explain the movement of charges and the divergence of leaves. 3

17. What are the safe practices to avoid damage by static electricity in a factory handling inflammable liquids? 3

OR

Explain why lightning is dangerous when we stand under a tree or hold a metal umbrella. 3

18. Explain the relationship between force and change in shape for rubber and clay, and which regains shape. 3

OR

Explain why the force of gravity between two small objects on a table is not noticeable. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Tara's mass is 40 kg. Taking the weight of 1 kg as about 10 N, she stands on a spring balance on the Earth. Her teacher says that on a planet where gravity is weaker the balance would show a smaller reading, but her mass would not change. 5
(i) Find Tara's weight on the Earth.
(ii) What is the SI unit of weight?
(iii) Which quantity changes with the place: mass or weight?
(iv) A 5 kg bag has a weight of about 50 N. Explain the difference between the mass and weight of the bag.

20. Compare three non-contact forces: gravitational, magnetic and electrostatic on nature, attraction/repulsion, materials affected and daily life examples. 5

OR

Explain the working of a simple compass and how it can be made using a magnet, a needle and a cork. 5

END OF QUESTION PAPER — Total: 40 Marks